

IS-305: AI Vision Based Kitting Solution for Warehousing



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Chapter 1: Introduction and Problem Statement

This chapter establishes the motivation, context, and engineering basis for the AEGIS (AI-Enhanced Guidance and Inspection System) project. It begins by characterising the manual kitting process and its associated error profile within a high-mix electronics connector manufacturing environment, before systematically analysing the root causes of kitting failures. Existing verification methods and commercial alternatives are evaluated, and their limitations in the target operational context are identified. The chapter concludes by defining stakeholder requirements, functional requirements, and the project scope that collectively frame the design and development effort.

1.1 Problem Introduction

TE Connectivity Ltd. is a global manufacturer of connectivity and sensor solutions, serving industrial, automotive, aerospace, medical, and commercial transportation sectors. With annual revenues of approximately USD 17.3 billion (FY2025) and manufacturing operations spanning 105 principal sites across the Americas, EMEA, and APAC regions, TE Connectivity operates at a scale at which marginal inefficiencies in production processes translate into significant financial and reputational consequences. The company's product catalogue encompasses more than 242 billion components annually, including connectors, sensors, relays, precision wire and cable, and application tooling, each produced across a portfolio of approximately 4,000 distinct stock-keeping units (SKUs).

Within TE Connectivity's assembly plants — including facilities in China, Poland, and Mexico — a critical mid-stream operation known as cell kitting is performed manually by ground operators. Kitting refers to the process of collecting and assembling a prescribed set of components into a single unit or package, in preparation for downstream assembly, shipment, or customer delivery. Each kitting order specifies a particular mix of SKUs and quantities, drawn from a subset of the approximately 800 to 900 commonly active SKUs. Orders are typically fulfilled in one to two hours, with an estimated 50 to 100 individual packages packed per order at each kitting station. Given a team size of four to six operators per kitting line, the cumulative throughput per shift is substantial.

The manual nature of this process introduces an inherent susceptibility to human error. Operators must repeatedly identify, retrieve, and pack correct components across a high-variety product range, under production time pressure and across extended shift durations. The primary error types observed are: the inclusion of incorrect components (wrong part), omission of required components (missing part), and packing of incorrect quantities (wrong quantity — either too few or too many). These errors are not immediately self-evident at the point of commission, as the current verification infrastructure consists solely of a digital postal scale used by operators to weigh each completed kit against an expected aggregate weight. No real-time feedback mechanism exists to alert operators during the act of kitting itself.

The consequences of kitting errors propagate across the supply chain. Errors that pass the scale check may be identified at downstream quality control inspections, at end-of-line audits, or — in the worst case — only upon delivery to the customer, necessitating replacement orders, rework, or scrapping of affected kits. TE Connectivity's own estimate places the Cost of Poor

Quality (COPQ) attributable to kitting errors at between USD 10,000 and USD 20,000 per kitting line per year, accounting for rework labour, component replacement, logistics costs associated with replacement shipments, and the residual risk of customer complaints and reputational damage. Given that this figure represents only the directly quantifiable cost for a single line, the aggregate impact across TE Connectivity's multi-plant, multi-line operations represents a substantially greater financial exposure.

The AEGIS project is initiated in direct response to this operational problem. Its primary objective is to develop a proof-of-concept, edge-deployed, multi-modal error detection system that can identify kitting errors — specifically wrong components, missing components, and incorrect quantities — at the point of packing, in real time, and in a manner that prompts corrective action from the operator before a defective kit is completed. The system is intended to augment, rather than replace, the manual kitting workflow, operating within the constraints of an existing manufacturing environment without requiring changes to workstation layout, bin configurations, or standard operating procedures.

1.2 Root Cause Analysis

A structured analysis of the contributing factors to kitting errors reveals that the problem is not attributable to a single systemic failure, but rather to an interaction of cognitive, ergonomic, procedural, and technological factors that collectively elevate error probability in high-mix manual assembly environments.

At the operator level, cognitive overload is a primary contributor. Kitting operators must simultaneously manage the order specification, identify components across multiple bins containing visually similar parts — connectors and electronic components frequently share form factors, colour schemes, or packaging styles — and maintain accurate counts across varying quantities per kit. This cognitive demand is compounded by production time pressure, which incentivises speed over deliberate verification. Fatigue, particularly during later stages of extended shifts, further degrades attentional accuracy and increases the likelihood of substitution and omission errors. Workstation ergonomics may also play a role: where bin layout is suboptimal or where components are stored in configurations that require repeated handling, the physical and cognitive demands of each kit increment.

At the procedural level, the verification mechanism in current use — a post-kit gravimetric check using a digital scale — is structurally insufficient for reliable error detection. This limitation warrants technical elaboration. The scale provides a single aggregate weight measurement of a completed kit, which is then compared against an expected total weight derived from the order specification. This approach is vulnerable in several respects. First, it cannot localise or diagnose an error: a weight discrepancy may correspond to a missing part, an excess part, a wrong part, or any combination thereof, without any information as to which component is implicated. Second, the measurement is subject to component weight tolerances: the unit mass of individual components ranges from approximately 0.5 grams to 1 kilogram, and manufacturing tolerances mean that individual parts may deviate from their nominal weight, introducing measurement uncertainty at fine granularity. Third, and most critically, the aggregate nature of the measurement creates the possibility of error cancellation: a deficit from one missing component may be arithmetically offset by a surplus from an excess component of

similar mass, producing a kit weight that appears within tolerance despite containing multiple errors. This failure mode is undetectable by the scale alone. Fourth, even where a weight discrepancy is detected, the scale offers no information to guide corrective action — the operator must re-verify the entire kit manually, consuming cycle time and adding process friction.

At the organisational level, quality control inspections downstream of the kitting cell provide a second line of defence, but their placement in the workflow means that errors are not intercepted at their point of origin. The latency between error commission and detection can span from minutes to days, depending on where in the production chain the defect is identified. Errors that reach the customer represent the highest-cost failure mode, incurring not only replacement and logistics costs but also the less-quantifiable impact on customer satisfaction and brand trust.

Synthesising these contributing factors, the root cause of the kitting error problem can be characterised as the absence of a real-time, part-specific verification mechanism at the operator workstation. Current tools are reactive rather than proactive: they detect aggregate outcomes rather than individual actions, they provide no in-process guidance to the operator, and they enforce no corrective behaviour before a defective kit is dispatched.

1.3 Current Solutions and Limitations

A range of verification and error-proofing approaches exists within the broader manufacturing and logistics sector. These can be grouped into five categories: manual process controls, barcode and RFID-based identification, vision-based inspection systems, gravimetric verification, and automated picking technologies. Each is examined below in the context of TE Connectivity's operational requirements.

Manual process controls — including paper checklists, supervisor sign-offs, and peer verification — represent the baseline approach. While requiring no capital investment, they do not eliminate human error; they merely introduce an additional human in the verification loop, which itself is subject to the same cognitive and fatigue-related failure modes as the original kitting act. These controls also add cycle time without providing real-time feedback.

Barcode and RFID scanning — where each component or bin is scanned at the point of handling — offers precise part-level identification and can enforce sequence compliance. However, the implementation burden in TE Connectivity's environment is substantial. With approximately 800 to 900 active SKUs and a high-mix order profile, maintaining up-to-date barcode or RFID labelling across all bins and components introduces significant setup and maintenance overhead. More critically, these systems require an additional handling step per component — the scan — which directly increases operator cycle time per kit and represents a productivity cost that may be disproportionate relative to the error rate it addresses. Operators in fast-paced kitting environments are also susceptible to scan-skipping behaviours, undermining system integrity.

Commercial machine vision systems — such as those offered by Cognex, Keyence, and similar providers — are capable of high-accuracy part identification and defect detection in controlled environments. However, they are generally optimised for fixed-configuration, single-product or narrow-product-range inspection tasks on automated production lines. In TE Connectivity's

high-mix kitting context, where the set of components changes with each order and bins may contain parts of widely varying form factors, the configuration and retraining effort for commercial systems would be considerable. The licensing and hardware cost of such systems is also typically high relative to the COPQ figures associated with the target problem, reducing the return on investment. Additionally, commercial systems are typically designed as standalone inspection stations rather than as operator-facing real-time guidance tools.

Pick-to-light systems — which illuminate the correct bin and prompt the operator with the required quantity — address the guidance problem effectively but require fixed bin configurations and dedicated hardware installation at each workstation position. They cannot verify that the correct component was retrieved from the illuminated bin, nor that the correct quantity was actually placed in the kit. They also impose rigid workstation layout constraints that may conflict with TE Connectivity's operational flexibility requirements.

Gravimetric verification — as currently employed at TE Connectivity — has been characterised in Section 1.2 as structurally insufficient for reliable error detection at the component level, owing to its aggregate measurement approach, susceptibility to error cancellation, and inability to provide diagnostic or corrective guidance.

A common limitation across all existing approaches is their inability to fulfil three simultaneous functions: predicting or detecting errors at the point of occurrence, diagnosing the specific nature of the error, and enforcing corrective action by the operator before the kit is completed. No single existing solution addresses all three functions in a form that is cost-effective, adaptable to a high-mix environment, and compatible with an unmodified manual kitting workflow.

1.4 Stakeholder Requirements

The AEGIS system serves a diverse set of stakeholders whose requirements must be identified and balanced in the system design. These stakeholders span the operational, managerial, technical, and commercial dimensions of TE Connectivity's manufacturing enterprise, as well as the academic context of the project.

Stakeholder	Role	Primary Requirements	Key Concerns
Ground Operator	Primary end-user of the system at the kitting workstation	Minimal disruption to kitting rhythm; clear and immediate error feedback; system must not add handling steps	Cycle time impact; ease of use; false alarm frequency
Production Supervisor	Oversees cell kitting lines and operator performance	Reliable error capture; visibility of operator compliance; minimal supervision overhead	System reliability; alert accuracy; ease of monitoring
Plant Production Manager	Responsible for line throughput and cost performance	Reduction in COPQ; no degradation of throughput; low total cost of ownership	ROI; system uptime; scalability across lines

Quality Engineer	Responsible for in-process and outgoing quality assurance	High detection rate for missing and wrong components; audit trail of kitting events	False negative rate; traceability; integration with QC processes
TE Connectivity (Organisation)	Problem owner and industrial partner	Deployable across China, Poland, and Mexico plants; adaptable to varied environments; cybersecurity and data sensitivity compliance	Generalisation across sites; IP and data protection
End Customer / Client	Recipient of kitted component orders	Correct components delivered in correct quantities; elimination of replacement orders	Order accuracy; delivery reliability
Maintenance / Plant Engineer	Responsible for system upkeep and configuration for new SKUs	Simple onboarding process for new components; minimal specialist expertise required for configuration	Onboarding time; calibration ease; hardware serviceability
AdvanTech (Hardware Partner)	Supplier of edge computing and HMI hardware used in the system	Hardware must be featured and validated within the prototype	Hardware compatibility; performance validation

Table 1.1: Stakeholder requirements summary.

Several tensions exist across stakeholder requirements that inform design trade-offs throughout the project. The ground operator's requirement for minimal cycle time impact is in tension with quality engineers' desire for high detection coverage; this trade-off is addressed by designing the system to operate passively during normal kitting and to intervene only upon detection of an anomaly. The production manager's cost sensitivity is addressed by positioning the system's total hardware cost below the lower bound of the annual per-line COPQ figure of USD 10,000, thereby ensuring a favourable return on investment within the first year of deployment.

1.5 Functional Requirements

The functional requirements of AEGIS are derived from the operational problem characterised in Sections 1.1 through 1.3 and from the stakeholder requirements defined in Section 1.4. They are presented below as discrete, verifiable requirements, distinguished from non-functional considerations where appropriate.

ID	Category	Requirement	Rationale
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FR-01	Component Identification	The system shall identify which component type is present in each active kitting bin at the operator workstation.	Enables part-specific rather than aggregate verification; prerequisite for wrong-part and missing-part detection.
FR-02	Wrong Part Detection	The system shall detect when a component placed into a kit does not correspond to the component specified in the active order.	Directly addresses the wrong-part error type identified as a primary failure mode.
FR-03	Missing Part Detection	The system shall detect when the kit is completed with one or more required components absent or below the specified quantity.	Missing parts represent the highest-priority error type due to their direct impact on customer order fulfilment.
FR-04	Quantity Verification	The system shall verify that the quantity of each component placed into the kit matches the quantity specified in the order.	Quantity errors (too few or too many) are not reliably detectable by gravimetric methods alone due to weight tolerance and cancellation effects.
FR-05	Real-Time Alerting	The system shall notify the operator of a detected error at the workstation display within a timeframe sufficient to enable corrective action prior to kit completion.	Proactive in-process notification is the defining capability absent from all current solutions; post-hoc detection is already partially addressed by downstream QC.
FR-06	Corrective Guidance	The system shall present actionable corrective guidance to the operator upon error detection, specifying the nature and location of the error.	Without specific guidance, operators cannot efficiently resolve errors; vague alerts increase cycle time and operator frustration.
FR-07	Order Integration	The system shall accept order specifications — including component identities, bin assignments, and required quantities — as inputs to define the verification context for each kit.	Verification must be order-specific; a part correct for one order may be incorrect for another.
FR-08	New Part Onboarding	The system shall support efficient onboarding of previously unseen component types without requiring specialist machine learning expertise from plant engineers.	TE Connectivity's 4,000-SKU catalogue and ongoing new product introductions necessitate a scalable and operationally

			accessible onboarding process.
FR-09	Environmental Robustness	The system shall operate correctly under varying ambient lighting conditions and shall not require modification of bin dimensions, colours, workstation layout, or operator workflow.	It is operationally infeasible to modify assembly line configurations to accommodate a verification system; the system must adapt to the environment.
FR-10	Supervisor Visibility	The system shall provide supervisory personnel with visibility of kitting status and error events across the workstation under their oversight.	Supervisors require situational awareness without being required to physically attend each workstation.
FR-11	Scalability	The system architecture shall support extension to multiple concurrent workstations without fundamental redesign of the per-station hardware or software stack.	Single-workstation deployment is the prototype scope; multi-workstation deployment is the intended production scenario.

Table 1.2: Functional requirements for the AEGIS system.

In addition to the functional requirements above, several non-functional considerations constrain the design. The system must not measurably increase the per-kit cycle time experienced by the operator under normal (error-free) kitting conditions, as any throughput reduction would need to be justified against the COPQ savings. The total hardware cost per workstation must be demonstrably below the lower bound of the annual per-line COPQ figure of USD 10,000 to support a positive return on investment within the first operating year. The system must be deployable and configurable by plant engineers without requiring involvement of external specialists for routine configuration and onboarding tasks. Data generated by the system must be handled in a manner consistent with TE Connectivity's data sensitivity requirements, with all processing performed on-device and no production data transmitted to external networks.

1.6 Value Proposition & Scope

1.6.1 Value Proposition

AEGIS addresses a well-defined operational gap: the absence of real-time, part-specific error detection and corrective guidance at the cell kitting workstation. The system's value is grounded in three distinct contributions relative to the current state at TE Connectivity.

First, AEGIS shifts the point of error detection upstream, from post-kit gravimetric verification or downstream QC inspection to the moment of component placement. This temporal shift fundamentally changes the cost profile of kitting errors: an error intercepted before kit

completion requires only a simple corrective action by the operator, whereas the same error detected downstream or post-delivery incurs rework labour, material replacement, logistics, and potential customer remediation costs. The COPQ figure of USD 10,000 to USD 20,000 per line per year represents the aggregate of these downstream costs; reducing the downstream error rate directly reduces this figure.

Second, the system employs complementary sensing modalities — computer vision for visual part identification and load cell-based gravimetric sensing for continuous quantity monitoring — to achieve a detection capability that neither modality can provide independently. Computer vision enables part-specific identification, distinguishing visually similar components that share similar masses, while gravimetric sensing provides a continuous, passive quantity signal that does not require the operator to perform additional handling or scanning steps. The integration of these two data streams addresses the diagnostic ambiguity inherent in single-modality approaches.

Third, the system is designed specifically for the high-mix, manual kitting context that commercial alternatives address poorly. Its edge-deployed architecture, order-driven verification context, and environment-agnostic operational requirements allow it to function across TE Connectivity's geographically and operationally diverse manufacturing sites without site-specific infrastructure investment.

1.6.2 Project Scope

The scope of the AEGIS project, as a proof-of-concept prototype, is precisely bounded to manage complexity within the constraints of available time, budget, and resources.

The following are within scope: the design, integration, and validation of a single-workstation AEGIS unit comprising computer vision and gravimetric sensing subsystems; the development of software for real-time component identification, error detection, and operator alerting; an operator-facing HMI display and a supervisor-facing display; a hardware architecture designed to support multi-workstation data aggregation, though multi-workstation physical deployment is not validated; and laboratory-based prototype validation using simulated kitting tasks representative of TE Connectivity's operational workflow.

The following are explicitly outside scope: deployment across multiple simultaneous physical workstations in a production environment; integration with TE Connectivity's enterprise resource planning (ERP) or manufacturing execution system (MES); automation of any component of the kitting process (the system is designed to work alongside manual operations); modification of existing standard operating procedures; and the handling of components that are opaque to camera-based identification due to packaging (e.g., sealed bags with no visible identifying features, where bin-level identification must substitute for part-level identification).

The prototype is validated against the generalised kitting workflow common across TE Connectivity's China, Poland, and Mexico facilities. Differences in specific workstation layouts,

bin configurations, and local operating practices are treated as environmental variables that the system is required to accommodate by design, consistent with the robustness requirements stated in FR-09. The system is not required to demonstrate cross-plant deployment in the prototype phase; rather, the architectural decisions made in the prototype are required to be demonstrably extensible to multi-plant deployment.

Chapter 2: Design Overview

2.1 Concept Overview

2.2 Design Selection

This section presents a morphological matrix of each feature and our options of approach towards it. It is split into two levels, 2.2.1 which covers the systemic decisions, while 2.2.2 covers the software and hardware selections

2.2.1 System Selection

Item	A	B	C	D	Reasoning
Worker Dashboard	Monitor	Physical Buttons	Audio / Voice Control		Provides intuitive clear instructions and feedback
Feedback Mechanism	Audio Alerts	Haptic Feedback	Visual Feedbacks		Non-intrusive and does not affect neighbouring working
Processing Architecture	Edge Ai PC	Raspberry Pi	Workstation Desktop	Micro Controller	Faster Processing and Industrial Grade
Mounting	Fixed Mount	Adjustable Arm	Rail Mounting		To minimise variables, and ensure hands are and bin visible
Item Identification	RFID	CV	Manual Initialization		SKUs too large for CV

					and RFID. Manual Init
Quantity Identification	Weight	CV	Manual Counter	Optical Gate	Weight database already established
Worker Intention	Weight	CV	Neural Implant	Optical Gate	Most Accurate and readily available models

2.2.2 Detailed Selection

Items	A	B	C	D	Reasoning
Monitor Application	Web Application	Desktop Application	Mobile Application	Terminal Based	
Edge Ai PC					
Hand Detection Model					
Bin Boundary Model					
Weight Sensors	Kitchen Scale	1x Load Cell	Human Weighing Scale	3x Load Cell	

2.2.3 Finite State Machine

Chapter 3: Prototype Development

Each section - Physical Frame, Computer Vision, UI, and Load cells were prototyped independently. Only when they fulfilled their minimum criteria alone was full integration testing done.

3.1 Physical Frame

3.2 Sensors

3.2.1 Computer Vision

3.2.2 Load Cells

As established in the Design Overview (Chapter 2), load cells were selected as the measurement method because no other approach could reliably identify components on a consistent basis. This decision was further supported by the availability of a TE component database containing reference weights and specifications for each item, allowing measured mass to be matched against known values for verification.

The initial design consisted of a simple cantilever platform, with the load cell secured by screws and the item rest was formed from bolts. However, the bolted arrangement proved unstable and required items to be placed directly onto the load cell, which was impractical for routine use.

The second iteration introduced a circular acrylic platform to improve stability. While this allowed a small bin to be seated securely, it remained prone to tipping when the bin's contents shifted to one side, displacing the centre of gravity beyond the platform's stable region. The resulting final design for small bins was a circular disk of [X] cm in diameter, which provided adequate stability for the expected load distribution.

Larger bins presented an additional challenge. Their greater footprint required a correspondingly wider platform, but single-point load cells maintain rated accuracy only within a specified distance from their centre; off-centre loading beyond this range introduces measurement error. A larger single-cell platform would therefore improve stability at the expense of accuracy.

To resolve this trade-off, a three-load-cell configuration was adopted. Distributing the load across three points ensured that the bin's centre of gravity always fell within the triangle formed by the cells, providing both mechanical stability and accurate readings. Three cells were chosen over four because a three-point arrangement guarantees simultaneous, full contact between all cells and the platform on any surface, whereas a four-cell layout is statically over-constrained and can leave one cell under-loaded if the surface is not perfectly flat.

Furthermore, the three-load-cell configuration allowed for scalability to fit the dimensions of any bins the factory uses by just ensuring that the 3 points of contact remained within the off center loading limit.

Finally, the system's weight constraints must account for load cell resolution. A load cell's accuracy is specified as a percentage of its rated maximum capacity, with the capacity indicated by its designation: a 1 kg load cell has a maximum capacity of 1 kg and a nominal error of ± 0.5 g, while a 5 kg load cell has a maximum capacity of 5 kg and a correspondingly larger nominal error of ± 2.5 g. As a result, a higher-capacity cell trades absolute resolution for a greater usable range. In our own testing, however, the 5 kg load cells arranged in the three-cell configuration achieved an error of ± 0.5 g, comparable to that of the 1 kg unit, as the distributed arrangement reduces the load borne by each individual cell.

Because all load cells share the same physical design and form factor, they are mechanically interchangeable, allowing the appropriate capacity to be selected according to the expected item weight without modification to the platform. The required capacity is therefore driven by two factors: the weight of the components being measured and the maximum permissible weight of the bin. By industry standards, a single bin may not exceed 25 kg, with a typical *"heavy" bin falling in the 10–15 kg range*, which informs the upper bound of the load cell capacity required for the larger platforms.

3.3 User Interface

3.3.1 Frontend

3.3.2 Backend

Chapter 4: Testing, Evaluation and Iterations

4.1 Physical Frame

4.2 Sensors

4.2.1 Computer Vision

4.2.1.1 Testing Framework

4.2.1.2 Evaluation Results

4.2.2 Load Cells

4.2.2.1 Testing Framework

4.2.2.2 Evaluation Results

4.3 User Interface

4.3.1 Frontend

4.3.1.1 Testing Framework

4.3.1.2 Evaluation Results

4.3.2 Backend

4.3.2.1 Testing Framework

4.3.2.2 Evaluation Results

Chapter 5: Conclusion

5.1

Chapter 6: Reflections and Lessons Learned

References

Appendix